

JNTU Kakinada

SUPPLEMENTARY EXAMINATION ? 2026

Database Management Systems ? Supplementary Examination, 2026

Subject Code: CS401 Semester: 4 Regulation: R23

Time: 3 Hours Max. Marks: 70

.....
Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
.....

SECTION A (Unit 1: ER Model and Relational Model)

1. Explain the ER model with a suitable example. Draw an ER diagram for a Library Management System and convert it into relational tables. [12 Marks]
2. What are the relational model concepts? Explain primary, candidate, composite and foreign keys with examples. [12 Marks]
3. Derive the expression related to ER Model and Relational Model and explain each term. [12 Marks]

SECTION B (Unit 2: SQL and Relational Algebra)

4. Write SQL queries for the given STUDENT?COURSE database covering joins, nested queries, aggregate functions, GROUP BY/HAVING and views. [12 Marks]
5. Explain the relational algebra operations ? selection, projection, union, intersection, join and division ? with examples. [12 Marks]
6. How is SQL and Relational Algebra implemented? Give an algorithm / procedure and explain. [12 Marks]

SECTION C (Unit 3: Normalization)

7. What is normalization? Explain 1NF, 2NF, 3NF and BCNF with examples and functional dependencies. [12 Marks]
8. Define Normalization. Explain its significance in Database Management Systems with a suitable example. [12 Marks]
9. Derive the expression related to Normalization and explain each term. [12 Marks]

SECTION D (Unit 4: Transactions and Concurrency Control)

10. Explain the ACID properties of transactions. What is serializability? Explain conflict serializability with an example. [12 Marks]
11. Describe the two-phase locking protocol and timestamp-based concurrency control protocol. [12 Marks]
12. Define Transactions and Concurrency Control. Explain its significance in Database Management Systems with a suitable example. [12 Marks]

SECTION E (Unit 5: File Organization and Indexing)

13. Explain the B+ tree index structure and its advantages over B-tree for database

indexing. [12 Marks]

14. Compare sequential, hashed and indexed file organization methods with respect to insertion, deletion and retrieval. [8 Marks]

15. Describe the working principle of File Organization and Indexing with necessary equations. [12 Marks]

*** End of Question Paper ***